

HOYBUKTMOEN KIRKENES, Norway

WMO: 010890

Lat: 69.726N

Lon: 29.893E

Elev: 86

StdP: 100.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.8	-23.8	-29.2	0.3	-26.4	-26.0	0.4	-23.6	14.4	-2.9	13.1	-3.7	3.2	140	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	23.1	15.1	20.3	14.5	18.1	13.2	16.3	21.3	14.9	19.2	13.7	17.5	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.1	18.3	12.8	9.3	16.9	11.7	8.6	15.5	46.0	21.2	42.0	19.0	38.6	17.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.6	9.4	DB	-29.3	26.4	2.4	2.9	-31.0	28.4	-32.5	30.1	-33.8	31.7	-35.6	33.8	(4)
(5)			WB	-29.4	18.5	2.4	2.7	-31.1	20.4	-32.6	22.0	-33.9	23.4	-35.6	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-10.3	-10.4	-6.1	-1.4	3.9	8.3	12.3	11.3	7.3	1.3	-4.8	-7.1	(6)
(7)		DBStd	9.14	6.63	6.63	4.82	3.86	3.75	3.61	3.72	3.26	3.24	3.84	5.05	5.80	(7)
(8)		HDD10.0	3684	628	571	501	343	194	76	16	21	93	269	443	529	(8)
(9)		HDD18.3	6542	887	804	759	593	448	302	191	219	331	528	693	787	(9)
(10)		CDD10.0	192	0	0	0	0	4	24	88	62	13	0	0	0	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0	(11)
(12)		CDH23.3	64	0	0	0	0	2	3	48	12	0	0	0	0	(12)
(13)		CDH26.7	9	0	0	0	0	0	0	8	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	4.9	5.0	5.2	4.5	4.3	4.3	3.8	3.6	4.2	4.6	4.7	5.1	(14)
(15)	Precipitation	PrecAvg	467	33	32	24	23	28	51	62	58	43	46	32	33	(15)
(16)		PrecMax	582	63	62	54	62	48	115	111	176	70	90	58	60	(16)
(17)		PrecMin	367	15	8	8	6	15	10	17	18	16	20	9	13	(17)
(18)		PrecStd	57	13	14	11	14	9	28	25	34	17	19	11	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	3.1	4.0	9.9	19.1	23.2	27.8	25.2	18.1	11.0	4.4	4.0	(19)
(20)			MCWB	1.3	1.7	1.8	6.1	11.3	14.8	18.0	15.7	12.9	8.5	3.3	2.2	(20)
(21)		2%	DB	1.0	1.4	2.6	6.5	15.0	19.7	24.1	21.8	15.8	9.2	3.0	2.2	(21)
(22)			MCWB	-0.1	0.3	0.9	3.5	8.9	12.9	15.8	15.1	11.7	7.5	2.1	1.0	(22)
(23)		5%	DB	-0.7	-0.1	1.1	4.9	12.1	16.9	21.6	19.0	13.9	7.8	2.0	1.1	(23)
(24)			MCWB	-1.5	-0.9	-0.3	2.4	7.7	11.5	15.1	14.0	10.9	6.4	1.1	0.2	(24)
(25)		10%	DB	-2.0	-1.6	0.0	3.6	9.7	14.2	19.0	16.8	12.1	6.4	1.1	0.0	(25)
(26)			MCWB	-2.6	-2.4	-1.2	1.7	6.2	9.8	14.1	13.1	9.9	5.2	0.4	-0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	1.9	2.0	5.9	12.3	16.0	18.7	17.7	13.8	9.0	3.5	2.4	(27)
(28)			MCDWB	2.5	2.9	3.4	9.1	17.5	20.6	24.7	22.8	16.9	10.4	4.1	3.7	(28)
(29)		2%	WB	0.0	0.4	1.0	3.7	9.5	13.7	17.0	15.8	12.4	7.7	2.2	1.2	(29)
(30)			MCDWB	0.8	1.3	2.3	5.9	14.0	17.8	22.6	19.7	14.9	8.9	2.7	2.1	(30)
(31)		5%	WB	-1.4	-0.9	-0.3	2.7	8.1	12.0	15.5	14.5	11.2	6.4	1.2	0.3	(31)
(32)			MCDWB	-0.7	-0.1	0.9	4.4	11.8	16.6	20.3	18.2	13.5	7.6	1.7	1.0	(32)
(33)		10%	WB	-2.7	-2.4	-1.2	1.9	6.6	10.2	14.2	13.3	10.1	5.2	0.4	-0.8	(33)
(34)			MCDWB	-2.1	-1.7	-0.1	3.2	9.3	13.8	18.6	16.4	11.9	6.2	0.9	0.0	(34)
(35)	Mean Daily Temperature Range	MDBR	6.5	6.3	6.0	5.7	5.7	6.0	6.8	6.2	5.2	3.7	4.9	5.8	(35)	
(36)		5% DB	MCDBR	6.0	5.4	5.2	6.1	10.5	10.9	11.9	10.8	7.7	4.5	4.2	4.9	(36)
(37)			MCWBR	5.6	5.0	4.5	4.1	6.0	5.4	5.4	5.1	4.5	3.6	3.9	4.6	(37)
(38)		5% WB	MCDBR	6.0	5.4	5.0	5.4	9.8	10.4	11.1	9.3	6.8	4.2	4.0	4.9	(38)
(39)			MCWBR	5.7	5.1	4.5	3.8	5.9	5.6	5.7	5.1	4.4	3.5	3.9	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.204	0.250	0.298	0.326	0.349	0.345	0.360	0.344	0.309	0.265	N/A	N/A	(40)	
(41)		taud	1.923	2.060	2.155	2.223	2.363	2.467	2.453	2.479	2.529	2.219	N/A	N/A	(41)	
(42)		Ebn at Noon	16	471	699	803	832	851	820	791	724	457	0	0	(42)	
(43)		Edh at Noon	3	41	77	99	98	91	90	78	56	35	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.01	0.33	1.62	3.34	4.43	4.76	4.37	2.99	1.59	0.46	0.00	0.00	(44)	
(45)		RadStd	0.00	0.04	0.10	0.24	0.45	0.38	0.46	0.30	0.13	0.06	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page